
CE 612: Theory of Plates and Shells

Course Code	CE 612
Course Name	Theory of Plates and Shells
L-T-P-C	3-0-0-3
Prerequisites	Theory of Elasticity, Advance Solid Mechanics (CE 557, ME 606)
Intended for	M.Tech./M.S./Ph.D./ B. Tech (3rd and 4th Year)
Distribution	Elective
Approval	33rd BoA (Jan 2020); 24th Senate (Feb 2020)

Course Contents

- Introduction to classical plate theory and governing equations, stress resultants, boundary conditions. [6 Lectures]
- Solution of bending of rectangular plates using Navier and Levy methods. [6 Lectures]
- Bending of circular plates. [6 Lectures]
- Vibration and buckling of rectangular and circular plates. [6 Lectures]
- Introduction to surfaces, Strain displacement relations in curvilinear coordinates, Classification of shells. [6 Lectures]
- Membrane theory of shells and governing equations, stress resultants, boundary conditions. [6 Lectures]
- Analysis of axisymmetric shells. [6 Lectures]
- Bending, buckling and vibration of circular cylindrical shells. [6 Lectures]

Text Books

1. S.P. Timoshenko and S. Woinowsky-Krieger, **Theory of Plates and Shells**, 2nd Edition, McGraw Hill, 2010.
2. W. Soedel, **Vibrations of Shells and Plates**, Marcel Dekker, 2004.
3. D.O. Brush and B.O. Almorth, **Buckling of Bars, Plates and Shells**, McGraw Hill, 1975.

References

1. H. Kraus, **Thin Elastic Shells**, John Wiley and Son, 1967.
2. E. Ventsel and T. Krauthammer, **Thin Plates and Shells Theory, Analysis, and Applications**, Marcel Dekker, 2001.
3. K. Chandrashekara, **Theory of Plates**, Universities Press, 2001.
4. J.N. Reddy, **Theory and Analysis of Elastic Plates and Shells**, CRC Press, 2006.